LIS 467 Web Development & Information Architecture

Assignment 1: Project Plan & Needs Assessment

1. Company Name: Agentic Curriculum Designers, Inc.

Logo:



2. Team members and roles (when working alone, imagine you're a web design company of 3-5 people; you can use fictitious names for other team members):
<Ambrose Choyce> - Designer
<Kameron Cole> - Programmer
<Jayne Sobczyk-Moore> - Designer

Reasons for choosing the number of members and assigning the roles this way:

We felt that three members was the perfect amount of people needed for a project of this scale and type. For our roles, we had discussed when initially signing up what roles each of us felt would fit us best based on our individual skill sets, and these reflect our own choices. Kameron has had a lot of experience with programming projects in the past, and both Ambrose and Jayne are more versed in the visual aspect of web design.

3. Project schedule / timeline

Oct 3	1st synchronous meeting: discussion on finalizing Assignment 1
Oct 5	Assignment 1 due
	Research: Mass STE Standards Research: Standards-based STEM projects Research: Science museum curation?
	Develop Information Architecture of the website.
	Create a wireframe to illustrate site layout.

	Construct a mood board that describes the styling of the website.
TBD	2nd synchronous meeting: review & finalize IA, wireframe, & moodboard
Oct 19	Assignment 2 due for peer review
Oct 26	Assignment 2 final due
	General work on the programming and design of the website
TBD	3rd synchronous meeting: review website work
TBD	4th synchronous meeting: create the URI/URL of website version created using an AI tool, start working on the video presentation
Dec 14	Final website due

4. Skills needed – and plans to acquire them

1. AI integration into interactive teacher/student webpage
    a. **Acquisition:** Research this topic through articles online and reviewing sample AIs similar to the one we plan to create.
2. Similarity/semantic search for webpages, with/without a database
    a. **Acquisition:** Research this topic through articles online and reviewing sample semantic searches similar to the one we plan to implement.
3. HTML5
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.
4. Node.js
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.
5. Javascript
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.

5. Site chosen – reasons for choice, and if you'll get in touch

The site we chose for this assignment is the Museum of Science (MOS) in Boston (https://www.mos.org/). The current MOS website lacks important background information on how their exhibits relate to specific STEM topics, nor how they facilitate learning connections for teachers, parents, or students. In order to create a unified picture of how a school assignment might incorporate an exhibit's teachings, we plan to connect teaching standards to exhibit information and external resources for educators. For students, exhibit information will be connected to external resources, while additional STEM activities will be connected to exhibit information for parents and other visitors. Additionally, we hope to incorporate a customized AI chatbot based on the user's

identity that will help them navigate and search the site easier. Once the project is completed, we hope to share it with the museum.

6. Audience / stakeholders

The MOS needs to attract interest from local families, and educators interested in the teaching materials they sell online as well as potential field trips. Educators need easy ways to compare lesson plans and activities with state mandated learning objectives. Fielding complicated questions from educators is time consuming and making the website more user friendly will help staff as well.

As such, the audience for this website will be teachers and parents who hope to teach their students or children more about STEM topics and wish to use the MOS for this purpose. An additional audience will be students doing research on STEM topics, either out of their own desire to learn or based on assignments given to them by their teachers/parents. The website will link additional resources about the exhibits and assist users in topic exploration.

7. Plan to gather requirements

1) Need to collect and download the https://www.doe.mass.edu/stem/ste/standards.html.
2) Investigate the interactive DESE Standards Navigator Tool.
3) May need to interview public school STEM teachers for design requirements.
4) Curate document seats geared to the student persona, that would aid in structure research assignments, per activity.

8. Mission/purpose

Mission Statement:

The mission of this website is to facilitate interactive learning and assist both educators and students in making connections with more intuitive searching techniques.

Design Statement:

Provide unity in the design of course curricula and education activities, utilizing the exhibits at the museum, but cross-referencing these with actual educational objectives of the State Massachusetts, particularly in STEM. The idea is: an educator wants to create an assignment that incorporates the museum experience, but aligned with specific STEM Objectives. For this, they would need additional information (the MOS site does not provide much information), a dynamic linking of the exhibit, the Objective, and extra information resources. Additionally, the creation of a chat bot for students and teachers linked to an AI model for open discussion.

9. Typical users

The typical users for this hypothetical website will be those hoping to learn more about STEM topics, with the different types of users being teachers, parents, and students.

10. Sample persona



Persona: Ms. Frizzle

- Teaches 5th grade science classes at a Boston public elementary school.
- Has been teaching for 15 years, always looking to update and improve her curriculum.
- Her science course has the budget for 2 full-day field trips per year. She must research and present these trips to the school for approval on top of her busy teaching schedule.
    - These field trips must fulfill required learning objectives & have coordinating lesson plans at the time of approval.
    - Field trips must be approved before the beginning of the school year (early September). She needs ample notice of planned exhibits.
- She is always looking for interactive activities for students & new ways to get them interested in learning.